

Week 4 — Probability Review, and the Markowitz Portfolio

AD7031 Optimization Under Uncertainty • Instructor: Prof. H. Park • Sep 19, 2026

1. Random variables

back to week 2's Section 2 table: the OR column had **uncertain data** ξ . today, the vocabulary — textbook notation (X, f) first, this course's notation $(\tilde{\xi})$ at the end.

Random variable

- **random experiment**: its outcome is not known in advance
- **sample space** Ω = the set of all possible outcomes

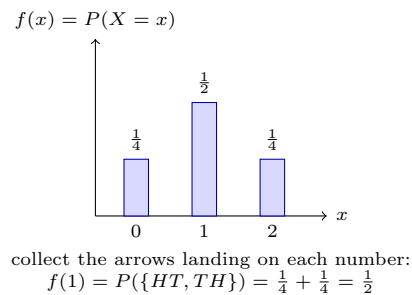
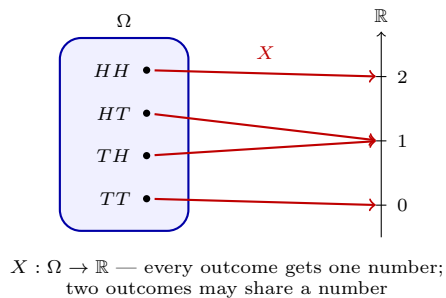
(week 2 wrote this S ; from here it is Ω , so that S can count *scenarios* below)

Def (*random variable*) a function that assigns a real number to each outcome: $X : \Omega \rightarrow \mathbb{R}$

capital X = the random variable, lowercase x = a value it takes

toss a coin twice: $\Omega = \{HH, HT, TH, TT\}$, each outcome has probability $\frac{1}{4}$

X = number of heads: $X(HH) = 2$, $X(HT) = 1$, $X(TH) = 1$, $X(TT) = 0$



- **discrete** X : countably many values — number of defective parts, number of arrivals, demand
- **continuous** X : values fill an interval — length, time, temperature, a return

2. Events, conditioning, independence

an **event** is a subset of the sample space, $A \subseteq \Omega$; $P(A)$ is the probability it occurs.

- $0 \leq P(A) \leq 1$, $P(\Omega) = 1$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ — subtract the overlap, or it is counted twice

Def (*conditional probability*) $P(A | B) = \frac{P(A \cap B)}{P(B)}$, defined when $P(B) > 0$

— B becomes the new sample space; divide by $P(B)$ so the weights add to 1 again

Def (*independent events*) A and B are independent if $P(A \cap B) = P(A)P(B)$

then $P(A | B) = \frac{P(A)P(B)}{P(B)} = P(A)$ — learning B tells you nothing about A

Thm (*law of total probability*) let $A_1, \dots, A_I$ **partition** Ω : $A_i \cap A_j = \emptyset$ for $i \neq j$, and $\bigcup_i A_i = \Omega$. then for any event B ,

$$P(B) = P(B \cap A_1) + \dots + P(B \cap A_I) = P(B | A_1)P(A_1) + \dots + P(B | A_I)P(A_I)$$

— split the world into cases, answer each case, weight by how likely the case is

two coins again. $A_1 = \text{first toss is } H$, $A_2 = \text{first toss is } T$ — a partition, each of probability $\frac{1}{2}$

take $B = \{X \geq 1\}$: $P(B | A_1) = 1$ (a head already landed), $P(B | A_2) = \frac{1}{2}$ (needs the second)

$P(B) = 1 \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{2} = \frac{3}{4}$ — the same $\frac{3}{4}$ the PMF gives in Section 3

takeaway conditioning is how a *two-stage* decision gets written down: act, watch which A_i occurred, act again. weeks 10–12 run on this.

3. Discrete random variables

Def (*probability mass function, PMF*) $f(x) = P(X = x)$, $f : \mathbb{R} \rightarrow [0, 1]$

(1) $f(x) \geq 0$ (2) $\sum_x f(x) = 1$ $P(X \in A) = \sum_{x \in A} f(x)$

two coins: $f(0) = \frac{1}{4}$, $f(1) = \frac{1}{2}$, $f(2) = \frac{1}{4}$, $P(X \geq 1) = f(1) + f(2) = \frac{3}{4}$

The same object, this course's way a discrete random variable is a **list of scenarios** with weights:

$\tilde{\xi}$ takes the values $\xi_1, \dots, \xi_S$ with $p_s = \mathbb{P}(\tilde{\xi} = \xi_s)$, $p_s \geq 0$, $\sum_{s \in [S]} p_s = 1$

here $[S] = \{1, \dots, S\}$ and s indexes *scenarios* — S of them, listed out

two coins: three scenarios $\xi_1 = 0$, $\xi_2 = 1$, $\xi_3 = 2$ with $p = (\frac{1}{4}, \frac{1}{2}, \frac{1}{4})$

same content as the PMF, different packaging. a PMF is a function; a scenario list is a *table* — and a solver can only be handed a table. every model from week 10 on is written this way.

Def (*mean, variance*)

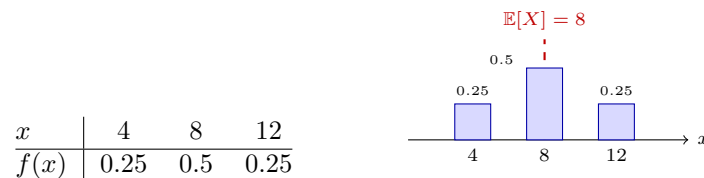
mean: $\mu = \mathbb{E}[X] = \sum_x x f(x)$

variance: $\sigma^2 = \text{Var}(X) = \sum_x (x - \mu)^2 f(x) = \mathbb{E}[X^2] - \mu^2$, standard deviation $\sigma = \sqrt{\text{Var}(X)}$

function of X : $\mathbb{E}[h(X)] = \sum_x h(x) f(x)$ $\leftarrow$ law of the unconscious statistician

$\mathbb{E}[aX + b] = a \mathbb{E}[X] + b$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$

Example 1 demand X for a spare part next month:



$\mathbb{E}[X] = 0.25 \cdot 4 + 0.5 \cdot 8 + 0.25 \cdot 12 = 8$

$\text{Var}(X) = 0.25 \cdot (4 - 8)^2 + 0.5 \cdot (8 - 8)^2 + 0.25 \cdot (12 - 8)^2 = 8$, $\sigma = 2\sqrt{2} \approx 2.83$

4. Continuous random variables

Def (*probability density function, PDF*) $f : \mathbb{R} \rightarrow \mathbb{R}_+$ with

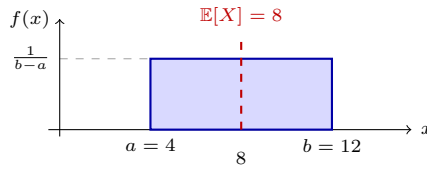
(1) $f(x) \geq 0$ (2) $\int_{-\infty}^{\infty} f(x) dx = 1$ (3) $P(a \leq X \leq b) = \int_a^b f(x) dx$ = area under f from a to b

$f(x)$ is *not* a probability: $P(X = x) = 0$ for every single x , and $f(x) > 1$ is allowed

Def (*mean, variance*) same as discrete — the sums become integrals:

$\mu = \mathbb{E}[X] = \int x f(x) dx$, $\sigma^2 = \text{Var}(X) = \int (x - \mu)^2 f(x) dx = \mathbb{E}[X^2] - \mu^2$, $\mathbb{E}[h(X)] = \int h(x) f(x) dx$

Example 2 $X \sim \text{Uniform}[a, b]$ — $f(x) = \frac{1}{b-a}$ on $[a, b]$, 0 outside



mean:

$$\mathbb{E}[X] = \int_a^b x \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

second moment — $h(x) = x^2$:

$$\mathbb{E}[X^2] = \int_a^b x^2 \frac{1}{b-a} dx = \frac{1}{b-a} \left[\frac{x^3}{3} \right]_a^b = \frac{b^3 - a^3}{3(b-a)} = \frac{a^2 + ab + b^2}{3}$$

variance — subtract, no new integral:

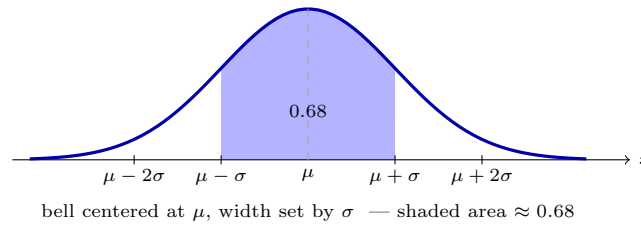
$$\text{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2 = \frac{a^2 + ab + b^2}{3} - \frac{(a+b)^2}{4} = \frac{(b-a)^2}{12}$$

with $a = 4, b = 12$: $\mathbb{E}[X] = 8, \quad \text{Var}(X) = \frac{64}{12} = \frac{16}{3}, \quad \sigma = \frac{4}{\sqrt{3}} \approx 2.31$

Normal distribution $X \sim \mathcal{N}(\mu, \sigma^2)$: $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad x \in \mathbb{R}$

$$\mathbb{E}[X] = \mu, \quad \text{Var}(X) = \sigma^2$$

$$P(\mu - \sigma \leq X \leq \mu + \sigma) \approx 0.68, \quad P(\mu - 2\sigma \leq X \leq \mu + 2\sigma) \approx 0.95$$



5. Expectation, in one formula

Sections 3 and 4 defined the mean twice — once a sum, once an integral. one definition covers both.

Def (expectation) for $\tilde{\xi} \sim \mathbb{P}$,

$$\mathbb{E}_{\mathbb{P}}[\tilde{\xi}] = \int_{\Omega} \xi \, \mathbb{P}(d\xi)$$

read $\mathbb{P}(d\xi)$ as *the probability sitting on a small piece $d\xi$* — the weight, whichever form it takes:

	the weight $\mathbb{P}(d\xi)$ is	and $\mathbb{E}_{\mathbb{P}}[\tilde{\xi}]$ reads
discrete	p_s , one per scenario ξ_s	$\sum_{s \in [S]} p_s \xi_s$
continuous	$p(\xi) d\xi$	$\int_{\Xi} \xi p(\xi) d\xi$

Def (expectation of a function) for $f: \mathbb{R}^k \rightarrow \mathbb{R}$ and a random vector $\tilde{\xi}$,

$$\mathbb{E}_{\mathbb{P}}[f(\tilde{\xi})] = \int_{\Omega} f(\xi) \, \mathbb{P}(d\xi) \quad \text{—} \quad \sum_{s \in [S]} p_s f(\xi_s) \quad \text{or} \quad \int_{\Xi} f(\xi) p(\xi) d\xi$$

this is the one the course actually needs: f will be a **cost**, and $\mathbb{E}[f(x, \tilde{\xi})]$ the objective.

variance is an expectation too, so it inherits the same two readings:

$$\text{Var}(\tilde{\xi}) = \mathbb{E}[(\tilde{\xi} - \mathbb{E}[\tilde{\xi}])^2], \quad \sigma(\tilde{\xi}) = \sqrt{\text{Var}(\tilde{\xi})}, \quad \text{Cov}(\tilde{\xi}_1, \tilde{\xi}_2) = \mathbb{E}[(\tilde{\xi}_1 - \mathbb{E}[\tilde{\xi}_1])(\tilde{\xi}_2 - \mathbb{E}[\tilde{\xi}_2])]$$

takeaway one definition, two computations. write $\mathbb{E}$ once and stop asking whether the data is discrete or continuous.

6. Two random variables

Def (joint PMF) $f_{XY}(x, y) = P(X = x, Y = y)$, $f_{XY} \geq 0$, $\sum_x \sum_y f_{XY}(x, y) = 1$

Def (marginal PMF) $f_X(x) = \sum_y f_{XY}(x, y)$, $f_Y(y) = \sum_x f_{XY}(x, y)$

— row sums and column sums, written in the *margins* of the table

Example 3 two assets, returns tomorrow: X, Y each either down (-0.10) or up ($+0.20$)

$f_{XY}(x, y)$	$y = -0.10$	$y = 0.20$	$f_X(x)$
$x = -0.10$	0.3	0.2	0.5
$x = 0.20$	0.2	0.3	0.5
$f_Y(y)$	0.5	0.5	1

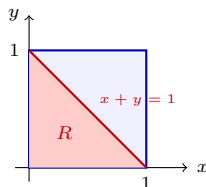
each asset alone: down or up w.p. $\frac{1}{2}$. together: same direction w.p. $0.3 + 0.3 = 0.6$

Def (joint PDF) $f_{XY} : \mathbb{R}^2 \rightarrow \mathbb{R}_+$, $\iint f_{XY}(x, y) dx dy = 1$, $P((X, Y) \in R) = \iint_R f_{XY}(x, y) dx dy$

— the **volume** under f_{XY} above the region R

marginal PDF: $f_X(x) = \int f_{XY}(x, y) dy$ — the sum becomes an integral

Example 4 (X, Y) uniform on the unit square: $f_{XY}(x, y) = 1$ on $[0, 1]^2$, 0 outside



$$P(X + Y \leq 1) = \iint_R 1 dx dy = \text{area of the triangle } R = \frac{1}{2}$$

$$\text{marginal: } f_X(x) = \int_0^1 1 dy = 1 \text{ on } [0, 1] \text{ — } X \text{ alone is Uniform}[0, 1]$$

Def (independence) X, Y are independent if $f_{XY}(x, y) = f_X(x) f_Y(y) \quad \forall x, y$ (PMF or PDF)

Example 3: $f_{XY}(-0.10, -0.10) = 0.3 \neq 0.5 \cdot 0.5$ — **not** independent

Example 4: $f_{XY}(x, y) = 1 = 1 \cdot 1$ — independent

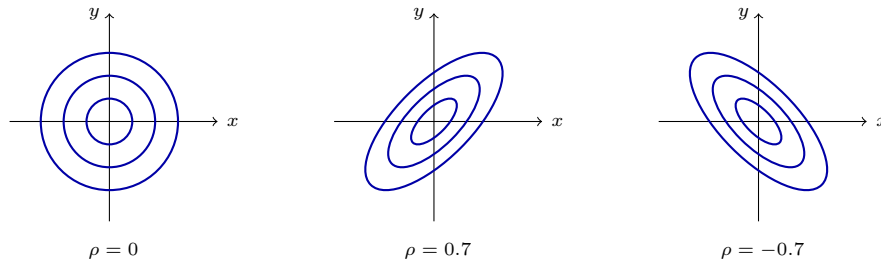
7. Covariance and correlation

Def (covariance, correlation)

$$\text{Cov}(X, Y) = \mathbb{E}[(X - \mu_X)(Y - \mu_Y)] = \mathbb{E}[XY] - \mu_X \mu_Y, \quad \text{Cov}(X, X) = \text{Var}(X)$$

$$\rho_{XY} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}, \quad -1 \leq \rho_{XY} \leq 1$$

$\rho > 0$: move together, $\rho < 0$: move opposite, independent $\Rightarrow \rho = 0$



contours of the joint PDF of two normal random variables, $\mu_X = \mu_Y = 0$, $\sigma_X = \sigma_Y = 1$

Example 3, continued (use the margins for μ and σ , all four cells for the covariance)

$$\mu_X = 0.5 \cdot (-0.10) + 0.5 \cdot 0.20 = 0.05, \quad \mu_Y = 0.05$$

$$\text{Var}(X) = 0.5 \cdot (-0.15)^2 + 0.5 \cdot (0.15)^2 = 0.0225, \quad \sigma_X = \sigma_Y = 0.15$$

$$\begin{aligned} \text{Cov}(X, Y) &= 0.3(-0.15)(-0.15) + 0.2(-0.15)(0.15) + 0.2(0.15)(-0.15) + 0.3(0.15)(0.15) \\ &= (0.3 - 0.2 - 0.2 + 0.3) \cdot 0.0225 = 0.0045 \end{aligned}$$

$$\rho_{XY} = \frac{0.0045}{0.15 \cdot 0.15} = 0.2 \quad \text{— the two assets tend to move together}$$

8. Linear combinations

$$\mathbb{E}[aX + bY] = a\mu_X + b\mu_Y \quad \text{— always, no independence needed}$$

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

n random variables $X = (X_1, \dots, X_n)$:

$$\text{mean vector } \mu = (\mathbb{E}[X_1], \dots, \mathbb{E}[X_n])^\top \in \mathbb{R}^n$$

covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$, $\Sigma_{ij} = \text{Cov}(X_i, X_j)$; diagonal $\Sigma_{ii} = \text{Var}(X_i)$; symmetric; always $\Sigma \succeq 0$

$$\text{for fixed } c \in \mathbb{R}^n: \quad \mathbb{E}[c^\top X] = c^\top \mu, \quad \text{Var}(c^\top X) = c^\top \Sigma c$$

$$n = 2, \quad c = (a, b): \quad c^\top \Sigma c = \begin{pmatrix} a & b \end{pmatrix} \begin{pmatrix} \text{Var}(X) & \text{Cov}(X, Y) \\ \text{Cov}(X, Y) & \text{Var}(Y) \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \quad \text{— Var}(aX + bY) \text{ again}$$

$$\text{Example 3: } \mu = \begin{pmatrix} 0.05 \\ 0.05 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.0225 & 0.0045 \\ 0.0045 & 0.0225 \end{pmatrix}$$

9. This course's notation, and the bridge

the decision is already x , so the uncertain data gets its own letter:

	textbook	this course
random variable	X	$\tilde{\xi}$, with distribution $\mathbb{P}$: $\tilde{\xi} \sim \mathbb{P}$
a value of it	x	ξ (a <i>realization</i>)
probability	$P(\cdot)$	$\mathbb{P}(\cdot)$
set of possible values	range of X	support Ξ (smallest closed set with $\mathbb{P}(\tilde{\xi} \in \Xi) = 1$)
discrete	values x , PMF $f(x)$	scenarios $\xi_1, \dots, \xi_S$ with probabilities $p_1, \dots, p_S$
continuous	PDF $f(x)$	density $p(\xi)$
mean, discrete	$\sum_x x f(x)$	$\mathbb{E}[\tilde{\xi}] = \sum_{s \in [S]} p_s \xi_s$
mean, continuous	$\int x f(x) dx$	$\mathbb{E}[\tilde{\xi}] = \int_{\Xi} \xi p(\xi) d\xi$
function of it	$\mathbb{E}[h(X)]$	$\mathbb{E}[f(\tilde{\xi})]$ (f = any function, e.g. a cost — not a PDF)
random vector	$X \in \mathbb{R}^k$, μ , Σ	$\tilde{\xi} \in \mathbb{R}^k$, μ , Σ

The bridge to optimization data random $\Rightarrow$ cost random: $f(x, \tilde{\xi})$ is a random variable, $\mathbb{E}[f(x, \tilde{\xi})]$ is a number.

$$\min_x \mathbb{E}[f(x, \tilde{\xi})]$$

takeaway every paradigm in this course differs in one place only: *how it collapses the randomness to a number.*

10. Running example: the Markowitz portfolio

a budget of 1, n assets. decide the split today; the returns are revealed tomorrow.

- **decision:** $x \in \mathbb{R}^n$, x_j = share of the budget in asset j
- **uncertain data:** $\tilde{\xi} = (\tilde{\xi}_1, \dots, \tilde{\xi}_n) \in \mathbb{R}^n$, $\tilde{\xi}_j$ = return of asset j — a random vector with mean μ , covariance matrix Σ
- **feasible set:** $X = \{x \in \mathbb{R}^n : \mathbf{1}^\top x = 1, x \geq 0\}$ ($\mathbf{1}$ = all-ones vector, $\mathbf{1}^\top x = x_1 + \dots + x_n$)

Portfolio return $\tilde{\xi}^\top x = x_1 \tilde{\xi}_1 + \dots + x_n \tilde{\xi}_n$ — a random variable

as a function: fix x , $f(\xi) = \xi^\top x$, $f: \mathbb{R}^n \rightarrow \mathbb{R}$ — Section 0's inner product, fed a random input

loss: $f(x, \tilde{\xi}) = -\tilde{\xi}^\top x$ (a gain is a negative loss)

Section 7's linear combinations with $c = x$: $\mathbb{E}[\tilde{\xi}^\top x] = \mu^\top x$, $\text{Var}(\tilde{\xi}^\top x) = x^\top \Sigma x$

Check on Example 3 $(\tilde{\xi}_1, \tilde{\xi}_2) = \text{Example 3's } (X, Y)$, half in each: $x = (\frac{1}{2}, \frac{1}{2})$

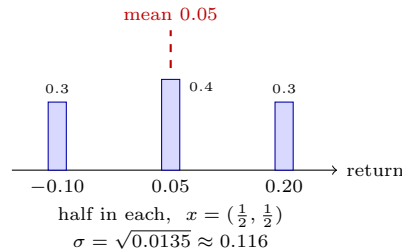
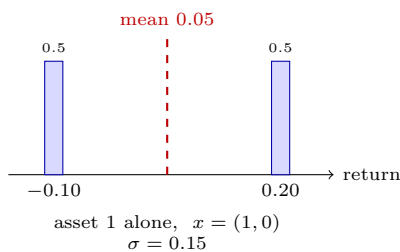
ξ_1	ξ_2	$\frac{1}{2}\xi_1 + \frac{1}{2}\xi_2$	probability	$\Rightarrow$ PMF of $\tilde{\xi}^\top x$:	value	probability	-0.10	0.05	0.20
-0.10	-0.10	-0.10	0.3						
-0.10	0.20	0.05	0.2				0.3	0.4	0.3
0.20	-0.10	0.05	0.2						
0.20	0.20	0.20	0.3						

from the PMF: $\mathbb{E}[\tilde{\xi}^\top x] = 0.3(-0.10) + 0.4(0.05) + 0.3(0.20) = 0.05$

$\text{Var}(\tilde{\xi}^\top x) = 0.3(-0.15)^2 + 0.4(0)^2 + 0.3(0.15)^2 = 0.0135$

from μ and Σ : $\mu^\top x = \frac{1}{2}(0.05) + \frac{1}{2}(0.05) = 0.05 \checkmark$

$x^\top \Sigma x = \frac{1}{4}(0.0225) + \frac{1}{4}(0.0225) + 2 \cdot \frac{1}{4}(0.0045) = 0.0135 \checkmark$



same mean, less spread: **diversification.** it comes from $\rho < 1$ — with $\rho = 1$, $x^\top \Sigma x$ would stay 0.0225.

The model (Markowitz, 1952) — least risk, subject to an expected-return target r :

$$\begin{aligned} \min_x \quad & x^\top \Sigma x && \text{variance of the return (risk)} \\ \text{s.t.} \quad & \mu^\top x \geq r && \text{expected return meets the target} \\ & \mathbf{1}^\top x = 1, \quad x \geq 0 \quad x \in X \end{aligned}$$

Section 5's QP with $Q = 2\Sigma$, $c = 0$; convex because $\Sigma \succeq 0$ (Section 6).

in the language of the bridge: the loss $-\tilde{\xi}^\top x$ is random; Markowitz collapses it with two numbers — its variance (objective) and its mean (constraint).

Data — the coding session $n = 3, r = 0.10$

asset j	expected return μ_j	variance Σ_{jj}	$\sigma_j = \sqrt{\Sigma_{jj}}$
bond	0.06	0.010	0.10
tech	0.10	0.040	0.20
energy	0.15	0.090	0.30

$$x = \begin{pmatrix} x_{\text{bond}} \\ x_{\text{tech}} \\ x_{\text{energy}} \end{pmatrix}, \quad \mu = \begin{pmatrix} 0.06 \\ 0.10 \\ 0.15 \end{pmatrix}, \quad \Sigma = \begin{pmatrix} 0.010 & 0.001 & 0 \\ 0.001 & 0.040 & 0.015 \\ 0 & 0.015 & 0.090 \end{pmatrix}$$

covariances: bond–tech 0.001, tech–energy 0.015, bond–energy 0

written out term by term — exactly the lines typed in the code:

$$\begin{aligned} \min \quad & 0.010 x_{\text{bond}}^2 + 0.040 x_{\text{tech}}^2 + 0.090 x_{\text{energy}}^2 + 2(0.001) x_{\text{bond}} x_{\text{tech}} + 2(0.015) x_{\text{tech}} x_{\text{energy}} \\ \text{s.t.} \quad & 0.06 x_{\text{bond}} + 0.10 x_{\text{tech}} + 0.15 x_{\text{energy}} \geq 0.10 \\ & x_{\text{bond}} + x_{\text{tech}} + x_{\text{energy}} \leq 1 \\ & x_{\text{bond}}, x_{\text{tech}}, x_{\text{energy}} \geq 0 \end{aligned}$$

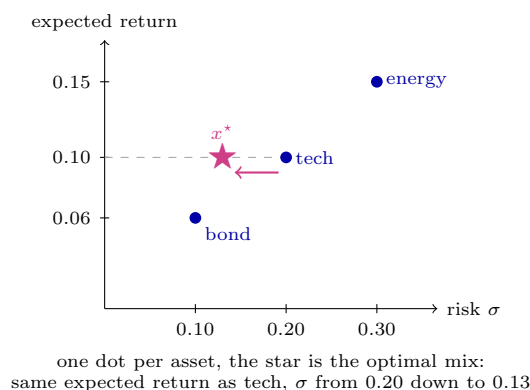
budget “ ≤ 1 ”: leftover sits in cash earning 0 — never helps, so the sum is 1 at the optimum

Solution

$$x^* \approx (0.39, 0.29, 0.31), \quad x^{*\top} \Sigma x^* \approx 0.0169, \quad \sigma \approx 0.130 \quad \mu^\top x^* = 0.10 \text{ — target met exactly}$$

all in tech: same expected return 0.10, but $\sigma = 0.20$. the mix cuts σ by about a third.

every asset gets a share — not a vertex (Section 5).



demo `week03_markowitz.ipynb` — Colab — build this model live from the skeleton, then compare with the completed cell.

demo `demos/markowitz_portfolio.py` — the same model as a plain script: prints the model, solves it, plots the weights x^* .

takeaway the running example: the loss $-\tilde{\xi}^\top x$ stays; each later model changes only how it is collapsed to a number — today variance, next CVaR and VaR.

next no class Sep 26. week 6 — duality: every LP comes with a shadow LP, and why $\max = \min$.